

Markscheme

November 2024

Chemistry

Higher level

Paper 2

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Question		Answers	Notes	Total
1.	a	$n(\text{HCl}) = 0.04000 \text{ dm}^3 \times 0.150 \text{ mol dm}^{-3} = 0.00600 / 6.00 \times 10^{-3} \text{ mol}$ □		1
	b	$3\text{HCl (aq)} + \text{Fe(OH)}_3 \text{ (s)} \longrightarrow \text{FeCl}_3 \text{ (aq)} + 3\text{H}_2\text{O (l)}$ □	Accept: ionic equation accepted	1
	c	$n(\text{HCl})_{\text{excess}} = \frac{1}{2} \times n(\text{NaOH}) = \frac{1}{2} \times (0.01840 \times 0.1205) = 1.109 \times 10^{-3} \text{ mol}$ □		1
	d	$n(\text{HCl})_{\text{reacted}} = 0.00600 - 0.001109 = 4.891 \times 10^{-3} \text{ mol}$ □		1
	e	$n(\text{Fe(OH)}_3) = \frac{1}{3} \times 4.891 \times 10^{-3} = 1.630 \times 10^{-3} \text{ mol}$ □ □ $m(\text{Fe(OH)}_3) = 1.630 \times 10^{-3} \times 106.87 = 0.1742 \text{ g}$ □	Award [[2]] for correct final answer	2
	f	$\% = (0.1742 / 1.50) \times 100 = 11.6 \%$ □	Answer must show 3 sig. figs.	1
	g	to reduce random errors / increase precision OR to check for reproducibility □	Accept: "to detect outliers"	1

Question		Answers	Notes	Total
2.	a	axes correctly labelled (fraction of particles vs kinetic energy) ☐ ☐ curve correct shape, starting at origin, not touching x-axis ☐ ☐ E_a (catalyst) < E_a (no catalyst) marked on x-axis ☐	M1: Accept speed/velocity for x-axis. M3: Ignore shading.	3
	b i	curve starts at origin, steeper gradient, reaches same maximum volume ☐		1
	b ii	rate decreases ☐ ☐ propanoic acid partially dissociated/ionized in solution, so lower $[H^+]$ ☐	Accept: "weak acid; higher pH"	2
	c	pH converts wide range of $[H^+]$ to simple logarithmic scale / avoids scientific notation ☐		1
	d i	A: CH_3CH_2COOH AND $CH_3CH_2COO^-$ ☐ ☐ B: $CH_3CH_2COO^-$ ☐		2
	d ii	$K_a = [H^+]^2 / 0.100$, so $[H^+] = 1.158 \times 10^{-3} \text{ mol dm}^{-3}$ ☐ ☐ pH = 2.94 ☐		2
	d iii	$CH_3CH_2COO^- (aq) + H_2O (l) \rightleftharpoons CH_3CH_2COOH (aq) + OH^- (aq)$ ☐		1
	d iv	less than 7 ☐		1
	e i	$SO_3 (g) + H_2O (l) \rightarrow H_2SO_4 (aq)$ ☐		1
	e ii	$H_2SO_4 (aq) + MgCO_3 (s) \rightarrow MgSO_4 (aq) + CO_2 (g) + H_2O (l)$ ☐		1

Question	Answers	Notes	Total
3.	a i four levels showing convergence at higher energy ☐		1
	a ii arrows (pointing down) from $n=3 \rightarrow n=2$ AND $n=4 \rightarrow n=2$ ☐		1
	a iii $E = h\nu = 6.63 \times 10^{-34} \times 6.17 \times 10^{14} = 4.09 \times 10^{-19} \text{ J}$ ☐		1
	a iv $\lambda = c/\nu = 3.00 \times 10^8 / 6.17 \times 10^{14} = 4.86 \times 10^{-7} \text{ m}$ ☐		1
	b i same shells/shielding AND nuclear charge increases across period ☐		1
	b ii Na^+ has 11 protons AND F^- has 9 protons ☐ ☐ isoelectronic; stronger nuclear pull in Na^+ ☐		2
	c i 4s: $\uparrow\downarrow$ 3d: $\uparrow\downarrow \uparrow\downarrow \uparrow\downarrow \uparrow\downarrow$ ☐	Accept any correct representation of Fe ($[\text{Ar}]3d^6 4s^2$)	1
	c ii Anode: $\text{Fe (s)} \rightarrow \text{Fe}^{2+} (\text{aq}) + 2\text{e}^-$ ☐ Cathode: $\text{Fe}^{2+} (\text{aq}) + 2\text{e}^- \rightarrow \text{Fe (s)}$ ☐	Award [1 max] if equations at wrong electrodes.	2
	c iii external wire/circuit AND from anode to cathode ☐		1
	c iv no change in concentration ☐	Do not accept "becomes colourless"	1
	c v oxygen / O_2 ☐		1
	d d and s orbitals close in energy, successive IEs increase gradually ☐ ☐ alkali metals: 2nd electron removed from much lower energy level ☐		2

Question		Answers	Notes	Total
4.	a	$\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{Fe}^{2+} (\text{aq}) \longrightarrow \text{Mn}^{2+} (\text{aq}) + 5\text{Fe}^{3+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l})$ □	Accept: $\rightleftharpoons$ for $\longrightarrow$	1
	b	$n = 5$ □ □ $E^0 = \Delta G / (-nF) = 419000 / (5 \times 96500) = 0.868 \text{ V}$ □		2
	c	$E^0(\text{MnO}_4^-/\text{Mn}^{2+}) = 0.868 + 0.77 = 1.64 \text{ V}$ □		1

Question	Answers	Notes	Total	
5.	a	bonds broken: $\text{N}_2 \rightarrow \text{N} + 3(\text{H-H}) = 945 + 3(436) = 2253 \text{ kJ}$ □ bonds formed: $6(\text{N-H}) = 6(391) = 2346 \text{ kJ}$ □ $\Delta H = 2253 - 2346 = -93 \text{ kJ}$ □	Award <i>[[3]]</i> for correct final answer Award <i>[[2 max]]</i> for -93 kJ without work	3
	b i	$\Delta H^\circ_f = 0$ by definition for elements in their standard states □		1
	b ii	$\Delta H^\circ = 2(-46.0) - [0 + 0] = -92.0 \text{ kJ}$ □		1
	b iii	bond enthalpies are average values across different molecules, not specific to these bonds □		1
	c	$\Delta S^\circ = 2(193) - [192 + 3(131)] = -199 \text{ J K}^{-1}$ □		1
	d	$\Delta G^\circ = -92.0 - 298(-199/1000) = -92.0 + 59.3 = -32.7 \text{ kJ}$ □		1
	e	$T = \Delta H / \Delta S = 92000 / 199 = 462 \text{ K}$ □	Do not award without calculation.	1

Question		Answers	Notes	Total
6.	a	Q_c : ratio of concentrations at any point, not necessarily at equilibrium K_c : ratio only at equilibrium		1
	b	$Q_c = (0.500)^2 / (1.00 \times 1.50^3) = 0.0741$ $Q_c < K_c$ so forward reaction favoured	<i>Do not award for direction without calculation.</i>	2
	c i	$[N_2O_4]$ decreases; endothermic forward reaction favoured at higher T		1
	c ii	$K_c(\text{step 1}) = [N_2O_2] / [NO]^2$, so $[N_2O_2] = K_c[NO]^2$ $\text{rate} = k[NO]^2[O_2]$		2
	d	$T_1 = 293 \text{ K}$, $T_2 = 303 \text{ K}$ $\ln(3) = E_a/R \times (1/293 - 1/303) \rightarrow E_a = 82.3 \text{ kJ mol}^{-1}$	<i>Award [[2]] for correct answer</i>	2

Question	Answers	Notes	Total
7.	a i	polar bond between H and halogen element □□electronegativity difference creates permanent dipole □	<i>Accept: "asymmetric charge distribution"</i> 2
	a ii	number of electrons increases from HCl to HI □□stronger London/dispersion/van der Waals forces □	<i>Accept: "larger surface area / molecular mass"</i> 2
	b	Electron domain geometry: tetrahedral □□Molecular geometry: trigonal pyramidal □	<i>Both marks from correct diagrams.</i> 2
	c i	O (double bond): formal charge = 0□O (single bond): formal charge = –1 □ □	2
	c ii	structure with all double bonds (three equivalent S=O) / no formal charges / minimises charge separation □	1
	d	Cl ₂ has weaker bond than F ₂ (longer, lower bond enthalpy) □□Cl ₂ requires longer wavelength (lower energy photon) □	2
	e	CO ₂ : non-polar, weak London/dispersion forces □□CS ₂ : larger molecule, stronger London/dispersion forces, more energy to separate □	2

Question	Answers	Notes	Total
8.	a	Physical: equal C–C bond lengths / bond order 1.5 for all C–C □□Chemical: undergoes substitution not addition / does not decolourize bromine water □	2
	b i	$3\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{OH} + \text{Cr}_2\text{O}_7^{2-} + 8\text{H}^+ \longrightarrow 3\text{CH}_3(\text{CH}_2)_2\text{CHO} + 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$ □	Award [1] reactants/products; [1] balance. 2
	b ii	Aldehyde: distilled from reaction mixture immediately as formed □□Carboxylic acid: heated under reflux to ensure complete oxidation □	2
	c i	$\text{MW}(\text{C}_3\text{H}_3\text{O}) = 55; 102/55 \longrightarrow \times 2 \longrightarrow \text{C}_6\text{H}_6\text{O}_2$ □	1
	c ii	A: C–H stretch (in aryl/alkyl group) □□B: C=O stretch (in ester/carbonyl) □	1
	c iii	Any two ester isomers of $\text{C}_6\text{H}_6\text{O}_2$ with phenyl group □ □	Award [[1 max]] for one correct ester structure 2
	c iv	$\text{C}_6\text{H}_5\text{COOCH}_3$ or appropriate structure based on NMR peaks □	1

Question		Answers	Notes	Total
9.	a i	Two ketone isomers of $C_6H_{12}O$ with $C=O$ not at chain end ☐ ☐ e.g. hexan-2-one AND hexan-3-one	<i>Accept condensed structural formulas.</i>	2
	a ii	A: hexan-2-one; peak at $m/z=43$ due to CH_3CO^+ ☐ ☐ B: hexan-3-one; peak at $m/z=57$ due to $C_2H_5CO^+$ ☐		2
	b i	heterolytic / heterolysis ☐		1
	b ii	polar protic ☐		1
	b iii	planar / trigonal carbocation drawn ☐ ☐ Shape: triangular planar ☐		2
	b iv	approximately 50% each ☐ ☐ nucleophile can attack from either face of the planar carbocation ☐	<i>Accept: "racemic mixture"</i>	2
	c	Stage 1: $CH_3C_6H_4NO_2 + 3Sn + 7H^+ \rightarrow CH_3C_6H_4NH_3^+ + 3Sn^{2+} + 2H_2O$ ☐ ☐ Stage 2: $CH_3C_6H_4NH_3^+ + OH^- \rightarrow CH_3C_6H_4NH_2 + H_2O$ ☐		2